

Практикум С7

108 marks of the archive, seven techniques, twelve tasks. Everything the archive measures with vectors from May 2021 to November 2025: the vector product itself, the area of a triangle and of a parallelogram, the volume of a pyramid, the identity that ties the two products together, the angle a line or a plane makes with a plane, the point of a line closest to a given point, and the shortest distance to a line.

О практикуме

Третий и последний практикум по векторам, и остаток после С5 и С6 оказался однородным: здесь векторами меряют. Площадь, объём, угол, расстояние — четыре величины, и все четыре берутся из векторного произведения. 48 баллов из 182 geometry.vectors (0.264) и 66 из 216 geometry.vectors_3d (0.3053). 114 баллов по корпусу, 108 рабочих — ноябрь 2023 Q8(a), (c) стоит дважды. 37 баллов Paper 1, 56 Paper 2, 21 Paper 3: единственный практикум по векторам, где Paper 3 весит.

Заданий

0

Раздел решений

включён

Интерактивная версия

kaggle.com/code/artemkonukhov/ib-math-praktikum-c7-measuring

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C7 — Measuring in space: the vector product

108 marks of the archive, seven techniques, twelve tasks. Everything the archive measures with vectors from May 2021 to November 2025: the vector product itself, the area of a triangle and of a parallelogram, the volume of a pyramid, the identity that ties the two products together, the angle a line or a plane makes with a plane, the point of a line closest to a given point, and the shortest distance to a line.

This is the third and last practicum on vectors. C5 took the points, the lines and the angles between them; C6 took the planes. What is left is one thing: **measuring**.

The one idea

A scalar product measures **how much two vectors agree**; a vector product

measures **how much they do not**.

$$\mathbf{u} \cdot \mathbf{v} = |\mathbf{u}||\mathbf{v}| \cos \theta \qquad |\mathbf{u} \times \mathbf{v}| = |\mathbf{u}||\mathbf{v}| \sin \theta$$

The first is a number, the second is a vector — perpendicular to both, and as long as the parallelogram they span is wide.

And what follows from it

the question asks	what it is
the area of a parallelogram	$ \mathbf{u} \times \mathbf{v} $, the two sides from one corner
the area of a triangle	half of it
the volume of a pyramid	a third of base times height — or a sixth of a determinant
$ \mathbf{u} $ from $\mathbf{u} \cdot \mathbf{v}$ and the area	$(\mathbf{u} \cdot \mathbf{v})^2 + \mathbf{u} \times \mathbf{v} ^2 = \mathbf{u} ^2 \mathbf{v} ^2$
the angle between two planes	the angle between their normals, taken acute
the angle between a line and a plane	what is left of a right angle with the normal
the shortest distance to a line	$ \overrightarrow{AP} \times \mathbf{v} / \mathbf{v} $

How the checks work

They do not know the answers. A figure is handed to them **the way the question names it**, and the check measures it:

```
verify_measure('4', q4, 'triangle', point_P,  
Q, R)  
verify_cross('2a', q2a, B - A, C - A)  
verify_distance('12b', q12b, l1, l2)
```

The first is *"is that the area of the triangle PQR?"* — the check builds the edges and measures. The second wants the vector product itself, not a multiple of it: unlike a normal, $\mathbf{u} \times \mathbf{v}$ has its own length and its own direction.

When you are wrong the check says **how**:

what you wrote	what the check says
$(2 - 3p, 2 + p, p^2 - 2p)$	the middle component of a vector product has the wrong sign
$12\sqrt{3}$ for the triangle	the half is missing: a triangle is half the parallelogram
1053 for the pyramid	that is the volume of the prism: a pyramid is a third of it
120° between two planes	that is the obtuse angle between the normals
0.639 for the angle with a plane	that is the angle with the normal
$\sqrt{54}$ for the distance between two lines	$ \mathbf{v} $ is not divided out

Order of work

level	what it means	tasks
	the product and what it measures	1-4
	the volume, the identity, the angle between planes	5-8
	angles with a line, the closest point, the distance	9-12

Every task is a real past-paper question, cited.

77 of the 114 marks are on a calculator paper, and here — unlike in C5 and C6 — it works. Three significant figures in an area, a volume, an angle or a distance are the ordinary answer of this topic, not the exception.

```

import sys
sys.path.append('..')          # from
practicum/geometry to practicum/kit/
import sympy as sp             # the escape
hatch: anything not in kit is in sp
from kit import *              # checks +
vec, cross, measure, plane, unknown

language('en')                # this
notebook is in English, and so are the checks

# cross(u, v) is the vector product; dot(u,
v) the scalar one; mag(v) the length.
# measure('triangle', A, B, C),
measure('parallelogram', A, B, C, D),
# measure('pyramid', S, A, B, C) – the figure
named by its vertices.
# angle(u, v) works for vectors, lines and
planes; distance(A, L) is the shortest one.
# verify_arc(label, got, r, u, v) is the way
along a sphere of radius r.

print('ready; sympy', sp.__version__)
u, v = vec(2, -1, 3), vec(1, 4, 0)
print('u × v:          ', cross(u, v))
print('perpendicular?  ', dot(cross(u, v),
u), dot(cross(u, v), v))
print('area of that triangle:',
measure('triangle', vec(0, 0, 0), u, v))

```

Map of the seven techniques

#	technique	you recognise it by	it reduces to
1	the vector product	<i>show that $\overrightarrow{AB} \times \overrightarrow{AC} = \dots$, find $\mathbf{a} \times \mathbf{p}$</i>	one determinant, three components
2	area	<i>the area of parallelogram ABCD, use a vector method to find the area</i>	$ \mathbf{u} \times \mathbf{v} $, halved for a triangle
3	volume	<i>find the volume of pyramid PQRS</i>	a third of base times height
4	the identity	<i>show that $(\mathbf{u} \cdot \mathbf{v})^2 + \mathbf{u} \times \mathbf{v} ^2 = \mathbf{u} ^2 \mathbf{v} ^2$, hence find $\mathbf{u}$</i>	$\cos^2 \theta + \sin^2 \theta = 1$
5	angles with a plane	<i>the acute angle between the planes, the angle between a line and a plane</i>	the normal, and an acute answer
6	the closest point	<i>the point on L nearest to the origin, find $\overrightarrow{PN} \cdot \overrightarrow{AB}$</i>	$\overrightarrow{PN} \cdot \mathbf{d} = 0$

#	technique	you recognise it by	it reduces to
7	the distance to a line	<i>the shortest distance from A to L_1, the minimum distance between l_1 and l_2</i>	$ \vec{AP} \times \mathbf{v} / \mathbf{v} $

Techniques 1–3 climb: a product, its length, a determinant of three. Technique 4 walks back to the scalar product and brings a length with it. Techniques 5–7 all say the same thing in three shapes — **the shortest way out is the perpendicular one.**

Part 1. The product and what it measures

Theory: the vector product

Two vectors in space have a third one standing across both of them:

$$\mathbf{u} \times \mathbf{v} = \begin{pmatrix} u_2 v_3 - u_3 v_2 \\ u_3 v_1 - u_1 v_3 \\ u_1 v_2 - u_2 v_1 \end{pmatrix}$$

For $\mathbf{u} = (2, -1, 3)$ and $\mathbf{v} = (1, 4, 0)$:

$$\mathbf{u} \times \mathbf{v} = \begin{pmatrix} (-1)(0) - (3)(4) \\ (3)(1) - (2)(0) \\ (2)(4) - (-1)(1) \end{pmatrix} = \begin{pmatrix} -12 \\ 3 \\ 9 \end{pmatrix}$$

Check it in two seconds. The product is perpendicular to both factors:
 $(-12, 3, 9) \cdot (2, -1, 3) = -24 - 3 + 27 = 0$ and
 $(-12, 3, 9) \cdot (1, 4, 0) = -12 + 12 = 0$. A sign slip almost never survives this test.

The middle component is $u_3 v_1 - u_1 v_3$, the other way round from its neighbours. Written as $u_1 v_3 - u_3 v_1$ it gives $(-12, -3, 9)$, and the check above

fails at once: $(-12, -3, 9) \cdot (2, -1, 3) = -24 + 3 + 27 = 6$.

Three things it is not.

- Not a number. $\mathbf{u} \cdot \mathbf{v} = 2 - 4 + 0 = -2$ is the scalar product; the vector product is a vector.
- Not the components multiplied one by one: $(2, -4, 0)$ is nothing.
- Not defined up to a factor. A normal of a plane may be scaled freely; here the length matters, because it is what measures:

$$|\mathbf{u} \times \mathbf{v}| = |\mathbf{u}||\mathbf{v}|\sin\theta$$

The order matters: $\mathbf{v} \times \mathbf{u} = -(\mathbf{u} \times \mathbf{v})$.

Task 1 — May 2025 TZ2 Paper 3 Q2(c), 5 marks

A sphere of radius 6 is centred at the origin O ; lengths are in thousands of kilometres. The North Pole P , a point N on the equator and a second equatorial point A have position vectors

$$\mathbf{p} = \begin{pmatrix} 0 \\ 0 \\ 6 \end{pmatrix}, \quad \mathbf{n} = \begin{pmatrix} 0 \\ 6 \\ 0 \end{pmatrix}, \quad \mathbf{a} = \begin{pmatrix} 6 \\ 0 \\ 0 \end{pmatrix}$$

P , N and A , and the arcs joining them, form a spherical triangle. **The angle at the vertex A is defined as the angle between the vectors $\mathbf{a} \times \mathbf{p}$ and $\mathbf{a} \times \mathbf{n}$.**

(i) Find the vector $\mathbf{a} \times \mathbf{p}$.

(ii) Show that the angle at the vertex A in the spherical triangle is 90° .

For (ii) enter the angle in degrees. Parts (a), (b), (d) belong to C1, C5 and C2; (e) and (f) are task 10.

```
q1_i = ...          # a x p
q1_ii = ...         # the angle at A, in degrees

a, p_pole, n = vec(6, 0, 0), vec(0, 0, 6),
vec(0, 6, 0)

verify_cross('1(i)', q1_i, a, p_pole)
verify_angle('1(ii)', q1_ii, cross(a,
p_pole), cross(a, n), deg=True)
```

Theory: the area

$|\mathbf{u} \times \mathbf{v}| = |\mathbf{u}||\mathbf{v}|\sin\theta$ is exactly base times height — **the area of the**

parallelogram on $\mathbf{u}$ and $\mathbf{v}$. A triangle is half of it.

Take $A(1, 0, 2)$, $B(3, 1, 2)$, $C(2, -1, 5)$. The two edges from A are $\overrightarrow{AB} = (2, 1, 0)$ and $\overrightarrow{AC} = (1, -1, 3)$, and

$$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{pmatrix} (1)(3) - (0)(-1) \\ (0)(1) - (2)(3) \\ (2)(-1) - (1)(1) \end{pmatrix} = \begin{pmatrix} 3 \\ -6 \\ -3 \end{pmatrix}$$

$$|\overrightarrow{AB} \times \overrightarrow{AC}| = \sqrt{9 + 36 + 9} = \sqrt{54} = 3\sqrt{6}$$

so the parallelogram on those edges has area $3\sqrt{6}$ and the triangle ABC has area $\frac{3\sqrt{6}}{2}$.

Two edges from one corner. In a parallelogram $ABCD$ the vertices go round, so the sides from A are $\overrightarrow{AB}$ and $\overrightarrow{AD}$ — not $\overrightarrow{AC}$, which is a diagonal.

$|\mathbf{u}||\mathbf{v}|$ **is not the area** unless the angle is right: $|\overrightarrow{AB}||\overrightarrow{AC}| = \sqrt{5} \cdot \sqrt{11} = \sqrt{55}$, and $\sqrt{54}$ is smaller — by exactly the factor $\sin \theta$.

With a letter in the points, the area is a function of it, and *smallest area* means the smallest value of that function.

Square first: minimise $|\mathbf{u} \times \mathbf{v}|^2$, which is a polynomial, and take the root at the very end.

Task 2 ● — **November 2023 TZ1**
Paper 2 Q8, 9 marks

Three points are given by $A(0, p, 2)$, $B(1, 1, 1)$ and $C(p, 0, 4)$, where p is a positive constant.

(a) Show that $\overrightarrow{AB} \times \overrightarrow{AC} = \begin{pmatrix} 2 - 3p \\ -2 - p \\ p^2 - 2p \end{pmatrix}$.

(b) Hence, find the smallest possible value of $|\overrightarrow{AB} \times \overrightarrow{AC}|^2$.

(c) Hence, find the smallest possible area of triangle ABC .

Give (b) and (c) to three significant figures.

```

p = symbols('p', positive=True)

q2a = ...          # AB × AC
q2b = ...          # the smallest value of |AB
× AC|2
q2c = ...          # the smallest area of ABC

A, B, C = vec(0, p, 2), vec(1, 1, 1), vec(p,
0, 4)
product = cross(B - A, C - A)

verify_cross('2a', q2a, B - A, C - A)
verify_optimum('2b', q2b, dot(product,
product), p, Interval(0, oo), 'min')
verify_optimum('2c', q2c, measure('triangle',
A, B, C), p, Interval(0, oo), 'min')

```

Task 3 ● — May 2025 TZ3 Paper 1 Q11(c), 4 marks

The points $A(1, -4, 0)$, $B(-3, -6, 2)$, $C(-1, -2, 4)$ and D form a parallelogram $ABCD$, where D is diagonally opposite B . (Part (a), technique 3 of C5, gives $D(3, 0, 2)$.)

(i) Given that $\overrightarrow{AB} \times \overrightarrow{AD} = m \begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix}$, where $m \in \mathbb{Z}^+$, find the value of m .

(ii) Hence, find the area of parallelogram $ABCD$.

Give (ii) exactly.

```
m = symbols('m')

q3_i = ...          # m
q3_ii = ...         # the area of ABCD

A, B, C, D = vec(1, -4, 0), vec(-3, -6, 2),
vec(-1, -2, 4), vec(3, 0, 2)

verify_find('3(i)', q3_i, m, [Eq(cross(B - A,
D - A), m*vec(-1, 1, -1))])
verify_measure('3(ii)', q3_ii,
'parallelogram', A, B, C, D, exact=True)
```

Task 4 — **November 2025 TZ1** **Paper 2 Q12(d), 5 marks**

The plane Π has Cartesian equation $4x + y - 3z = 18$ (*part (b), technique 2 of C6*). It meets the coordinate axes at $P(4.5, 0, 0)$, $Q(0, q, 0)$ and $R(0, 0, r)$, where $q = 18$ and $r = -6$ (*part (c), technique 1 of C6*).

(d) Use a vector method to find the area of the triangle PQR .

Three significant figures.

```
q4 = ...          # the area of PQR

point_P, Q, R = vec(Rational(9, 2), 0, 0),
vec(0, 18, 0), vec(0, 0, -6)

verify_measure('4', q4, 'triangle', point_P,
Q, R)
```

Part 2. The volume, the identity, the angle

Theory: the volume of a pyramid

A pyramid on a triangular base is a **third of base times height**:

$$V = \frac{1}{3} S_{\text{base}} \cdot h$$

The base area is technique 2. The height is the distance from the apex to the **plane** of the base — measured along the normal, not to a corner.

Take the triangle $A(1, 0, 2)$, $B(3, 1, 2)$, $C(2, -1, 5)$ again and the apex $S(1, 1, 6)$. The three edges from S are

$$\overrightarrow{SA} = \begin{pmatrix} 0 \\ -1 \\ -4 \end{pmatrix},$$

$$\overrightarrow{SB} = \begin{pmatrix} 2 \\ 0 \\ -4 \end{pmatrix},$$

$$\overrightarrow{SC} = \begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix}$$

and the shortest road to the volume is one determinant:

$$V = \frac{1}{6} \left| \det \begin{pmatrix} 0 & -1 & -4 \\ 2 & 0 & -4 \\ 1 & -2 & -1 \end{pmatrix} \right| = \frac{18}{6} = 3$$

The two routes are the same thing. The determinant is the volume of the **parallelepiped** on the three edges; a pyramid is a sixth of it, because the base is half a parallelogram and the height enters with a third.

what you computed	what it is
the determinant	the parallelepiped: six pyramids
base $\times$ height	the prism: three pyramids
$\frac{1}{2} \mathbf{u} \times \mathbf{v} \cdot h$	twice the pyramid — the $\frac{1}{3}$ is missing

If the apex sits on a normal line $\mathbf{r} = \mathbf{m} + \gamma \mathbf{n}$ at a known γ , the height is $|\gamma| |\mathbf{n}|$ — not $|\gamma|$.

Task 5 — **November 2025 TZ1**
Paper 2 Q12(g), 4 marks

Continuing task 4: the line L_3 is normal to Π and passes through $(1, 8, -2)$, so $\mathbf{r}_3 = \begin{pmatrix} 1 \\ 8 \\ -2 \end{pmatrix} + \gamma \begin{pmatrix} -4 \\ -1 \\ 3 \end{pmatrix}$. The point $S(-11, 5, 7)$ lies on L_3 at $\gamma = 3$ (*parts (e), (f), techniques 1 and 6 of C5-C6*).

(g) Hence, find the volume of pyramid $PQRS$.

The base PQR is the triangle of task 4. Give the answer exactly — it is a whole number.

```
q5 = ...           # the volume of PQRS

point_P, Q, R = vec(Rational(9, 2), 0, 0),
vec(0, 18, 0), vec(0, 0, -6)
S = vec(-11, 5, 7)

verify_measure('5', q5, 'pyramid', S,
point_P, Q, R, exact=True)
```

Theory: the identity that joins the two products

The two products see the same angle from two sides:

$$\mathbf{u} \cdot \mathbf{v} = |\mathbf{u}||\mathbf{v}| \cos \theta, \quad |\mathbf{u} \times \mathbf{v}| = |\mathbf{u}||\mathbf{v}| \sin \theta$$

Square both and add. The Pythagorean identity does the rest:

$$\begin{aligned} (\mathbf{u} \cdot \mathbf{v})^2 + |\mathbf{u} \times \mathbf{v}|^2 &= |\mathbf{u}|^2 |\mathbf{v}|^2 (\cos^2 \theta + \sin^2 \theta) \\ &= |\mathbf{u}|^2 |\mathbf{v}|^2 \end{aligned}$$

The same statement written the other way is $|\mathbf{u} \times \mathbf{v}|^2 = |\mathbf{u}|^2 |\mathbf{v}|^2 - (\mathbf{u} \cdot \mathbf{v})^2$, and one line takes you from either to the other.

What it is for. It gives a length that nobody handed you. If $\mathbf{a} \cdot \mathbf{b} = 12$, $|\mathbf{a} \times \mathbf{b}| = 5$ and $|\mathbf{b}| = 13$, then

$$\begin{aligned} 12^2 + 5^2 &= |\mathbf{a}|^2 \cdot 13^2 \\ &\implies \\ |\mathbf{a}|^2 &= \frac{169}{169} = 1 \\ &\implies \\ |\mathbf{a}| &= 1 \end{aligned}$$

Three habits are worth having here.

- The scalar product carries $\cos$, the vector product $\sin$. Swapping them is the commonest slip on the page.
- **Both terms are squared.** $\mathbf{u} \cdot \mathbf{v} + |\mathbf{u} \times \mathbf{v}|$ is not $|\mathbf{u}||\mathbf{v}|$.
- The identity gives $|\mathbf{u}|^2$; the question usually wants $|\mathbf{u}|$.

A Cartesian proof — expand both sides in components — is allowed, but the markscheme warns that a half-finished one scores nothing.

Task 6 — **May 2021 TZ2 Paper 1 Q5, 4 marks**

Given any two non-zero vectors $\mathbf{a}$ and $\mathbf{b}$, show that $|\mathbf{a} \times \mathbf{b}|^2 = |\mathbf{a}|^2|\mathbf{b}|^2 - (\mathbf{a} \cdot \mathbf{b})^2$.

The proof is two lines on paper. Enter its two ends: the left-hand side and the last term of the right-hand side, each written with $A = |\mathbf{a}|$, $B = |\mathbf{b}|$ and θ — and then read the identity off $\sin^2 \theta = 1 - \cos^2 \theta$. Use θ for θ .

```

A, B, th = symbols('A B th')

q6_i = ...          #  $|a \times b|^2$ , written with A,
B and th
q6_ii = ...         #  $(a \cdot b)^2$ , written with A,
B and th

a = vec(*symbols('a1 a2 a3'))          #
any two vectors, in components
b = vec(*symbols('b1 b2 b3'))
letters = {A: mag(a), B: mag(b), th: angle(a,
b)}

verify_formula('6(i)', q6_i, dot(cross(a, b),
cross(a, b)), letters)
verify_formula('6(ii)', q6_ii, dot(a, b)**2,
letters)

```

Task 7 — May 2024 TZ1 Paper 2 Q12(a), (b), 15 marks

Consider the non-zero vectors $\mathbf{u}$ and $\mathbf{v}$, and let θ be the angle between them.

(a) Using the definitions of $\mathbf{u} \cdot \mathbf{v}$ and $\mathbf{u} \times \mathbf{v}$ in terms of $|\mathbf{u}|$, $|\mathbf{v}|$ and θ , show that $(\mathbf{u} \cdot \mathbf{v})^2 + |\mathbf{u} \times \mathbf{v}|^2 = |\mathbf{u}|^2 |\mathbf{v}|^2$.

A triangle ABC has vertices $A(0, 1, 2)$, $B(p, q, 3)$ and $C(3, 2, 1)$, with $p, q \in \mathbb{R}$. The vectors are $\mathbf{u} = \overrightarrow{AB}$ and $\mathbf{v} = \overrightarrow{AC}$. It is given that $\mathbf{u} \cdot \mathbf{v} = 3$ and that the area of triangle ABC is $\sqrt{6}$.

(b) (i) Find the value of $|\mathbf{u} \times \mathbf{v}|$. **(ii)** Hence, or otherwise, find the value of $|\mathbf{u}|$. **(iii)** Hence, or otherwise, find the possible values of p and the corresponding values of q .

For (a) enter the left-hand side written with $U = |\mathbf{u}|$, $V = |\mathbf{v}|$ and th — before you simplify it. For (b)(iii) enter both pairs, as $[[p, q], [p, q]]$. Part (c) is the task on the clock.

```

U, V, th = symbols('U V th')
p, q = unknown('p'), unknown('q') #
the two the question asks for

q7a = ... #  $(u \cdot v)^2 + |u \times v|^2$ ,
written with U, V and th
q7b_i = ... #  $|u \times v|$ 
q7b_ii = ... #  $|u|$ 
q7b_iii = [...] # both pairs: [[p, q], [p,
q]]

one = vec(*symbols('m1 m2 m3')) #
any two vectors, for part (a)
two = vec(*symbols('n1 n2 n3'))
verify_formula('7a', q7a, dot(one, two)**2 +
dot(cross(one, two), cross(one, two)),
{U: mag(one), V: mag(two), th:
angle(one, two)})

A, B, C = vec(0, 1, 2), vec(p, q, 3), vec(3,
2, 1)
u, v = B - A, C - A
given = [Eq(measure('triangle', A, B, C),
sqrt(6)), Eq(dot(u, v), 3)]

verify_find('7b(i)', q7b_i, mag(cross(u, v)),
given)
verify_find('7b(ii)', q7b_ii, mag(u), given)
verify_find('7b(iii)', q7b_iii, [p, q],
given)

```

Theory: angles with a plane

A plane has no direction of its own — it has a normal, and every angle with a

plane is computed through it.

Two planes. The angle between them is the angle between the normals. Take $2x - y + 2z = 5$ and $x + 2y + 2z = 1$: the normals are $(2, -1, 2)$ and $(1, 2, 2)$,

$$\begin{aligned}\cos \theta &= \frac{|(2, -1, 2) \cdot (1, 2, 2)|}{3 \cdot 3} = \frac{|2 - 2 + 4|}{9} = \frac{4}{9} \\ &\implies \\ \theta &= 63.6^\circ\end{aligned}$$

The bars matter: without them a negative scalar product gives the obtuse angle, and the question almost always asks for the acute one. If your answer came out above 90° , subtract it from 180° .

A line and a plane. Here the normal points the wrong way, and the angle you want is what is left of a right angle:

$$\sin \alpha = \frac{|\mathbf{d} \cdot \mathbf{n}|}{|\mathbf{d}||\mathbf{n}|}$$

For the line with direction $(1, 2, 2)$ and the plane $x - y + z = 4$: $\sin \alpha = \frac{|1 - 2 + 2|}{3\sqrt{3}} = \frac{1}{3\sqrt{3}}$, so $\alpha = 11.1^\circ$.

Sine for a line and a plane, cosine for two planes. One formula, two names, because in the first case the normal is 90° off the thing you measure.

On a sphere the same rule appears in disguise: the angle between two arcs is the angle between the planes of their great circles, that is between $\mathbf{a} \times \mathbf{p}$ and $\mathbf{a} \times \mathbf{n}$ — vector products again. And an arc of angle θ on a sphere of radius r has length $r\theta$, with θ **in radians**.

Task 8 — **May 2025 TZ1 Paper 1 Q11(a) and May 2025 TZ3 Paper 1 Q11(e), 9 marks**

(a) (TZ1) The plane P_1 has equation $x + 2y + z = 0$ and the plane P_2 has equation $x - y - 2z = 0$. The acute angle between P_1 and P_2 is θ . Show that $\theta = 60^\circ$.

(b) (TZ3) The plane Π_1 has equation $-x + y - z = -5$ and the plane Π_2 has equation $5x + y - 7z = 1$. The acute angle between Π_1 and Π_2 is θ . Show that $\cos \theta = \frac{1}{5}$.

In (a) enter the angle in degrees; in (b) enter the cosine, exactly.

```
q8a = ...          #  $\theta$  in degrees
q8b = ...          #  $\cos \theta$ 

P1 = plane(Eq(x + 2*y + z, 0), name='P1')
P2 = plane(Eq(x - y - 2*z, 0), name='P2')
Pi1 = plane(Eq(-x + y - z, -5), name='Pi1')
Pi2 = plane(Eq(5*x + y - 7*z, 1), name='Pi2')

verify_angle('8a', q8a, P1, P2, deg=True)
verify_angle('8b', q8b, Pi1, Pi2,
cosine=True, exact=True)
```

● Part 3. The hard angles, and where the perpendicular lands

The first two tasks here are technique 5 again — the same normal, the same acute answer — on the two questions where the archive makes it hard: a letter inside the plane, and a triangle drawn on a sphere. Then the last two techniques, and both say the same thing: **the shortest way out is the perpendicular one.**

Task 9 ● — *May 2023 TZ1 Paper 2 Q8, 7 marks*

The angle between a line and a plane is α , where $0 < \alpha < \frac{\pi}{2}$. The equation of the line is

$$\frac{x-1}{3} = \frac{y+2}{2} = 5-z$$

and the equation of the plane is $4x + (\cos \alpha)y + (\sin \alpha)z = 1$.

Find the value of α .

Three significant figures, in radians — the markscheme gives $A0$ for degrees. The equation in α is one for the calculator.

```
alpha = symbols('alpha', positive=True)

q9 = ...          #  $\alpha$  in radians

L = cartesian((x - 1)/3, (y + 2)/2, 5 - z)
Pi = plane(Eq(4*x + cos(alpha)*y +
sin(alpha)*z, 1))

verify_find('9', q9, alpha, [Eq(angle(L, Pi),
alpha)])
```

Task 10 — May 2025 TZ2 Paper 3 Q2(e), (f), 11 marks

Continuing task 1, on the same sphere of radius 6 (thousands of kilometres), centred at O , with the North Pole $\mathbf{p} = (0, 0, 6)$:

Moscow M has $\mathbf{m} = \begin{pmatrix} 0 \\ 6 \cos \theta \\ 6 \sin \theta \end{pmatrix}$ where $\theta = 57.3^\circ$, and Bogotá B has $\mathbf{b} = \begin{pmatrix} 6 \sin 120^\circ \\ 6 \cos 120^\circ \\ 0 \end{pmatrix}$.

(e) Find the shortest distance from Bogotá to Moscow on the sphere.

(f) The bearing from B to M is the angle at the vertex B in the spherical triangle BMP — that is, by the method of part (c), the angle between $\mathbf{b} \times \mathbf{m}$ and $\mathbf{b} \times \mathbf{p}$. Find the bearing from Bogotá to Moscow.

Give (e) in thousands of kilometres and (f) in degrees, both to three significant figures.

```
q10_i = ...      # the distance along the
sphere, in thousands of km
q10_ii = ...     # the bearing, in degrees

theta = 57.3*pi/180                                #
57.3 degrees, from part (d)
b = vec(6*sin(2*pi/3), 6*cos(2*pi/3), 0) #
120 degrees west of Nairobi
m = vec(0, 6*cos(theta), 6*sin(theta))
p_pole = vec(0, 0, 6)

verify_arc('10(e)', q10_i, 6, b, m)
verify_angle('10(f)', q10_ii, cross(b, m),
cross(b, p_pole), deg=True)
```

Theory: the closest point on a line

Every point of a line is $\mathbf{c} + \mu\mathbf{d}$. The one closest to a given point P is the one where the segment to it **crosses the line at a right angle**:

$$\overrightarrow{PN} \cdot \mathbf{d} = 0$$

That is one linear equation in one unknown, and it always has exactly one root.

Take $P(6, 1, 3)$ and $L : \mathbf{r} = (1, 0, 2) + \mu(2, 1, -2)$. The general point is $N(1 + 2\mu, \mu, 2 - 2\mu)$, so

$$\begin{aligned}\overrightarrow{PN} &= \begin{pmatrix} 2\mu - 5 \\ \mu - 1 \\ -2\mu - 1 \end{pmatrix}, & \overrightarrow{PN} \cdot \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix} \\ &= 9\mu - 9\end{aligned}$$

Setting it to zero gives $\mu = 1$ and $N(3, 1, 0)$.

Three warnings.

- The answer is the **point**, not μ . Put μ back into the line.

- The scalar product is taken with the **direction**, not with the position vector of N — unless the given point is the origin, where the two happen to coincide. "*Nearest to the origin*" is the one case where $\overrightarrow{ON} \cdot \mathbf{d} = 0$ is the whole condition.
- $\overrightarrow{PN} = \overrightarrow{ON} - \overrightarrow{OP}$, end minus start. The other way round flips the sign of every term and sends μ the wrong way.

Minimising $|\overrightarrow{PN}|^2$ by calculus gives the same μ ; the markscheme accepts both, and the dot product is shorter.

Task 11 ● — **November 2025 TZ3**
Paper 1 Q10(d), (e) and May 2023
TZ2 Paper 2 Q6(b), 12 marks

(a) (November 2025 TZ3) The point $P(-1, 1, -13)$ lies on L_1 . The line L_2 passes through $A(2, -4, 2)$ and $B(7, -6, 1)$, so $\mathbf{s} = \begin{pmatrix} 2 \\ -4 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 5 \\ -2 \\ -1 \end{pmatrix}$, and the point N lies on L_2 . Find $\overrightarrow{PN} \cdot \overrightarrow{AB}$ in terms of μ .

(b) Given that N is the point on L_2 that lies closest to P , find the coordinates of N .

(c) (May 2023 TZ2) L is the line of intersection of $\Pi_1 : 2x - y + 2z = 6$ and $\Pi_2 : 4x + 3y - z = 2$, and a vector equation of it is $\mathbf{r} = \begin{pmatrix} 0 \\ 2 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -2 \\ -2 \end{pmatrix}$ (part (a), technique 4 of C6). Find the coordinates of the point P on L that is nearest to the origin.

In (a) use μ for μ ; give (b) and (c) exactly.

```

q11a = ...      # PN · AB, in terms of mu
q11b = ...      # N
q11c = ...      # the point of L nearest to
the origin

point_P, A, B = vec(-1, 1, -13), vec(2, -4,
2), vec(7, -6, 1)
L2 = line(A + mu*(B - A), mu, name='L2')
N = unknown('N', 3)

verify_find('l1(a)', q11a, dot(L2.at(mu) -
point_P, B - A))
verify_find('l1(b)', q11b, N, [on(N, L2),
perpendicular(N - point_P, L2)])

L = line(vec(0, 2, 4) + lam*vec(1, -2, -2),
lam, name='L')
closest = unknown('N', 3)

verify_find('l1(c)', q11c, closest,
[on(closest, L), perpendicular(closest, L)])

```

Theory: the distance to a line

The shortest distance from a point to a line is the length of the perpendicular — and the vector product measures it directly:

$$d = \frac{|\overrightarrow{AP} \times \mathbf{v}|}{|\mathbf{v}|}$$

where A is **any** point of the line and $\mathbf{v}$ its direction. The numerator is $|\overrightarrow{AP}||\mathbf{v}|\sin\theta$; dividing by $|\mathbf{v}|$ leaves $|\overrightarrow{AP}|\sin\theta$, which is exactly the height of P above the line.

From the origin to $\mathbf{r} = (1, 2, 2) + \lambda(0, 1, 0)$:
 $\overrightarrow{AP} = (-1, -2, -2)$ and

$$\frac{|(-1, -2, -2) \times (0, 1, 0)|}{|(0, 1, 0)|} = \frac{|(2, 0, -1)|}{1} = \sqrt{5}$$

Dividing by $|\mathbf{v}|$ is not optional. It is the single most common loss of a mark here: $|\overrightarrow{AP} \times \mathbf{v}|$ on its own is the area of a parallelogram, not a length.

Two parallel lines. Pick a point on each; the distance between the lines is the distance from one of them to the other line. For $l_1 : \mathbf{r} = \lambda(1, 1, 1)$ and $l_2 : \mathbf{r} = (1, 0, 0) + \mu(2, 2, 2)$ the directions are multiples, so the lines are parallel, and

$$\begin{aligned}
 d &= \frac{|(1, 0, 0) \times (1, 1, 1)|}{|(1, 1, 1)|} \\
 &= \frac{|(0, -1, 1)|}{\sqrt{3}} \\
 &= \frac{\sqrt{2}}{\sqrt{3}} \\
 &= \frac{\sqrt{6}}{3}
 \end{aligned}$$

Check that they are parallel first. If they are not, the two lines either meet — distance zero — or are skew, and the archive has never asked for that.

Two other routes reach the same number and the markscheme takes them: if the angle θ between $\overrightarrow{AP}$ and the line is known, $d = |\overrightarrow{AP}| \sin \theta$; and minimising $|\overrightarrow{AB}|^2$ over the parameter, or setting $\overrightarrow{AB} \cdot \mathbf{v} = 0$, gives the foot of the perpendicular and then the distance.

Task 12  — **May 2023 TZ1 Paper 1 Q12(c) and May 2021 TZ2 Paper 1 Q8(b), 9 marks**

(a) (May 2023 TZ1) Two lines L_1 and L_2 meet at P , and $A(2t, 8, 3)$ lies on L_2 . The

acute angle between the lines is $\frac{\pi}{3}$, the direction vector of L_1 is $(1, 1, 0)$, and $\overrightarrow{PA} = (2t, 0, 3 + t)$. Parts (a) and (b) (*technique 5 of C5*) give $t = 3$, so $P(0, 8, -3)$ and $A(6, 8, 3)$. Hence or otherwise, find the shortest distance from A to L_1 .

(b) (*May 2021 TZ2*) The lines l_1 and l_2 have vector equations

$$\begin{aligned} l_1 : \mathbf{r}_1 &= \begin{pmatrix} 3 \\ 2 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -2 \\ 2 \end{pmatrix} & l_2 : \mathbf{r}_2 \\ &= \begin{pmatrix} 2 \\ 0 \\ 4 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \end{aligned}$$

Part (a) (*technique 7 of C5*) shows that they do not intersect. Find the minimum distance between l_1 and l_2 .

Both answers are exact.

```
q12a = ...          # the shortest distance from
A to L1
q12b = ...          # the minimum distance
between l1 and l2

L1 = line(vec(0, 8, -3) + lam*vec(1, 1, 0),
lam, name='L1')
A = vec(6, 8, 3)
verify_distance('l2(a)', q12a, A, L1,
exact=True)

l1 = line(vec(3, 2, -1) + lam*vec(2, -2, 2),
lam, name='l1')
l2 = line(vec(2, 0, 4) + mu*vec(1, -1, 1),
mu, name='l2')
verify_distance('l2(b)', q12b, l1, l2,
exact=True)
```

Trainer: name the technique in five seconds

Twelve openings. Do not compute anything — say only **which move you would make first**.

code	technique
cross	the vector product itself: compute it, or check it
area	the area of a triangle or a parallelogram
volume	the volume of a pyramid
identity	$(\mathbf{u} \cdot \mathbf{v})^2 + \mathbf{u} \times \mathbf{v} ^2 = \mathbf{u} ^2 \mathbf{v} ^2$
angle	an angle with a plane, through its normal
closest	the point of a line closest to a given point
distance	the shortest distance to a line

1. Find $(1, 0, -2) \times (3, 1, 1)$.
2. Find the area of the triangle with vertices $(0, 0, 0)$, $(1, 2, 0)$, $(0, 3, 4)$.
3. A pyramid has base $(0, 0, 0)$, $(2, 0, 0)$, $(0, 3, 0)$ and apex $(0, 0, 5)$. Find its volume.
4. Given $\mathbf{u} \cdot \mathbf{v} = 6$, $|\mathbf{u} \times \mathbf{v}| = 8$ and $|\mathbf{v}| = 2$, find $|\mathbf{u}|$.
5. Find the acute angle between $x + y = 1$ and $x - z = 4$.
6. Find the point of $\mathbf{r} = \lambda(1, 2, 2)$ closest to $(3, 0, 0)$.

7. Find the distance from $(1, 1, 1)$ to the line $\mathbf{r} = \mu(0, 0, 1)$.
8. Find the area of the parallelogram with sides $(2, 1, 0)$ and $(0, 1, 3)$ from one corner.
9. Find the angle the line $\mathbf{r} = \lambda(1, 1, 1)$ makes with the plane $z = 0$.
0. Show that $(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{a} = 0$ for all $\mathbf{a}$ and $\mathbf{b}$.
1. Two parallel lines pass through $(0, 0, 0)$ and $(1, 1, 0)$ with direction $(0, 0, 1)$. How far apart are they?
2. Show that $|\mathbf{a} \times \mathbf{b}| \leq |\mathbf{a}||\mathbf{b}|$.

```

answers = {
    1: '', 2: '', 3: '', 4: '', 5: '', 6: '',
    7: '', 8: '', 9: '', 10: '', 11: '', 12:
    '',
}

trigger_check(answers, {1: 'b3986952b145', 2:
'4a91ee5f0106', 3: '62d7a6b1211d', 4:
'689f6a627384', 5: '49387897bee2', 6:
'2b327541ac3d', 7: '63e6ce93e1bc', 8:
'4a91ee5f0106', 9: '49387897bee2', 10:
'b3986952b145', 11: '63e6ce93e1bc', 12:
'689f6a627384'})

```

On the clock — May 2024 TZ1
Paper 2 Q12(c), 5 marks

Five marks, seven minutes. Calculator allowed, no hints.

A triangle ABC has vertices $A(0, 1, 2)$, $B(1, 2, 3)$ and $C(3, 2, 1)$, and $\mathbf{u} = \overrightarrow{AB}$, $\mathbf{v} = \overrightarrow{AC}$. Consider a new point D , and let $\mathbf{w} = \overrightarrow{CD}$.

It is given that $\mathbf{u} \cdot \mathbf{w} = \mathbf{v} \cdot \mathbf{w} = 0$ and that the area of triangle ACD is 5 square units.

Find the possible vectors for $\mathbf{w}$.

Enter both, as a list of two vectors.

Attempt log

date	time	result

```
qt = [...]          # both possible w

A, B, C = vec(0, 1, 2), vec(1, 2, 3), vec(3,
2, 1)
u, v = B - A, C - A
w = unknown('w', 3)
D = C + w

verify_find('timer', qt, w, [perpendicular(w,
u), perpendicular(w, v),
Eq(measure('triangle', A, C, D), 5)])
```



Solutions

Work these only after you have your own answer, or you are reading, not practising.

$$\mathbf{1} \quad \mathbf{(i)} \quad \mathbf{a} \times \mathbf{p} = (6, 0, 0) \times (0, 0, 6) = \begin{pmatrix} 0 \cdot 6 - 0 \cdot 0 \\ 0 \cdot 0 - 6 \cdot 6 \\ 6 \cdot 0 - 0 \cdot 0 \end{pmatrix} = \boxed{\begin{pmatrix} 0 \\ -36 \\ 0 \end{pmatrix}}$$

$\mathbf{1} \quad \mathbf{(ii)} \quad \mathbf{a} \times \mathbf{n} = (6, 0, 0) \times (0, 6, 0) = (0, 0, 36)$,
and $(0, -36, 0) \cdot (0, 0, 36) = 0$, so the angle is $\boxed{90^\circ}$.

$\mathbf{2} \quad \mathbf{(a)} \quad \overrightarrow{AB} = (1, 1 - p, -1)$ and $\overrightarrow{AC} = (p, -p, 2)$
, so

$$\begin{aligned} \overrightarrow{AB} \times \overrightarrow{AC} &= \begin{pmatrix} (1-p)(2) - (-1)(-p) \\ (-1)(p) - (1)(2) \\ (1)(-p) - (1-p)(p) \end{pmatrix} \\ &= \boxed{\begin{pmatrix} 2 - 3p \\ -2 - p \\ p^2 - 2p \end{pmatrix}} \end{aligned}$$

2 (b) $(2 - 3p)^2 + (2 + p)^2 + (p^2 - 2p)^2 = p^4 - 4p^3 + 14p^2 - 8p + 8$. Its derivative $4p^3 - 12p^2 + 28p - 8$ vanishes at $p = 0.3264\dots$, and the value there is $\boxed{6.75}$ ($6.75257\dots$).

2 (c) Area $= \frac{1}{2}|\overrightarrow{AB} \times \overrightarrow{AC}| = \frac{1}{2}\sqrt{6.75257\dots} = \boxed{1.30}$ units².

3 (i) $\overrightarrow{AB} = (-4, -2, 2)$, $\overrightarrow{AD} = (2, 4, 2)$, and $\overrightarrow{AB} \times \overrightarrow{AD} = (-4 - 8, 4 + 8, -16 + 4) = (-12, 12, -12) = 12(-1, 1, -1)$, so $\boxed{m = 12}$.

3 (ii) The area is that length: $12|(-1, 1, -1)| = \boxed{12\sqrt{3}} \approx 20.8$.

4 $\overrightarrow{PQ} = (-4.5, 18, 0)$ and $\overrightarrow{PR} = (-4.5, 0, -6)$, so $\overrightarrow{PQ} \times \overrightarrow{PR} = (-108, -27, 81)$ and

$$\begin{aligned}\text{area} &= \frac{1}{2}\sqrt{108^2 + 27^2 + 81^2} \\ &= \frac{1}{2}\sqrt{18954} \\ &= \boxed{68.8}\end{aligned}$$

5 S is at $\gamma = 3$ on the normal line, so the height is $3|(-4, -1, 3)| = 3\sqrt{26} = \sqrt{234}$, and

$$V = \frac{1}{3} \cdot \frac{1}{2} \sqrt{18954} \cdot \sqrt{234} = \boxed{351}$$

The determinant route is shorter: the edges from S are $(15.5, -5, -7)$, $(11, 13, -7)$, $(11, -5, -13)$, whose determinant is -2106 , and $2106/6 = 351$.

6 (i) $|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}||\mathbf{b}| \sin \theta$, so

$$\boxed{|\mathbf{a} \times \mathbf{b}|^2 = A^2 B^2 \sin^2 \theta}.$$

6 (ii) $\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos \theta$, so

$$\boxed{(\mathbf{a} \cdot \mathbf{b})^2 = A^2 B^2 \cos^2 \theta}. \text{ Then}$$

$$\begin{aligned} A^2 B^2 \sin^2 \theta &= A^2 B^2 (1 - \cos^2 \theta) \\ &= A^2 B^2 - A^2 B^2 \cos^2 \theta \\ &= |\mathbf{a}|^2 |\mathbf{b}|^2 - (\mathbf{a} \cdot \mathbf{b})^2 \end{aligned}$$

7 (a) $\boxed{U^2 V^2 \cos^2 \theta + U^2 V^2 \sin^2 \theta} = U^2 V^2 (\cos^2 \theta + \sin^2 \theta) = U^2 V^2$.

7 (b)(i) Area $= \frac{1}{2} |\mathbf{u} \times \mathbf{v}| = \sqrt{6}$, so $|\mathbf{u} \times \mathbf{v}| = \boxed{2\sqrt{6}} \approx 4.90$.

7 (b)(ii) $\mathbf{v} = (3, 1, -1)$, $|\mathbf{v}|^2 = 11$, and the identity gives $3^2 + (2\sqrt{6})^2 = |\mathbf{u}|^2 \cdot 11$, that is $33 = 11|\mathbf{u}|^2$ and $|\mathbf{u}| = \boxed{\sqrt{3}} \approx 1.73$.

7 (b)(iii) $\mathbf{u} = (p, q - 1, 1)$. From $\mathbf{u} \cdot \mathbf{v} = 3$: $3p + q - 1 - 1 = 3$, so $q = 5 - 3p$. From $|\mathbf{u}|^2 = 3$: $p^2 + (q - 1)^2 + 1 = 3$. Substituting, $p^2 + (4 - 3p)^2 = 2$, that is $10p^2 - 24p + 14 = 0$ and $5p^2 - 12p + 7 = 0$:

$$\boxed{p = 1, q = 2} \quad \text{or} \quad \boxed{p = \frac{7}{5}, q = \frac{4}{5}}$$

8 (a) The normals are $(1, 2, 1)$ and $(1, -1, -2)$, both of length $\sqrt{6}$, and their scalar product is $1 - 2 - 2 = -3$:

$$\cos \theta = \frac{|-3|}{6} = \frac{1}{2} \implies \boxed{\theta = 60^\circ}$$

Without the bars $\cos \theta = -\frac{1}{2}$ and $\theta = 120^\circ$; the acute angle is $180^\circ - 120^\circ$.

8 (b) The normals are $(-1, 1, -1)$ and $(5, 1, -7)$, of lengths $\sqrt{3}$ and $\sqrt{75} = 5\sqrt{3}$, with scalar product $-5 + 1 + 7 = 3$:

$$\cos \theta = \frac{3}{\sqrt{3} \cdot 5\sqrt{3}} = \frac{3}{15} = \boxed{\frac{1}{5}}$$

9 The direction of the line is $(3, 2, -1)$ and the normal of the plane is $(4, \cos \alpha, \sin \alpha)$, of lengths $\sqrt{14}$ and $\sqrt{16+1} = \sqrt{17}$. The angle between them is $\frac{\pi}{2} - \alpha$, so

$$\cos\left(\frac{\pi}{2} - \alpha\right) = \sin \alpha = \frac{12 + 2 \cos \alpha - \sin \alpha}{\sqrt{14}\sqrt{17}}$$

that is $12 + 2 \cos \alpha - \sin \alpha = \sqrt{238} \sin \alpha$, and the calculator gives $\boxed{\alpha = 0.932}$ (0.932389...). In degrees it is 54.4° — and the markscheme gives A0 for that.

10 (e) $\mathbf{b} = (5.196, -3, 0)$ and $\mathbf{m} = (0, 3.240, 5.050)$, both of length 6:

$$\cos \widehat{BOM} = \frac{\mathbf{b} \cdot \mathbf{m}}{36} = \frac{-9.721}{36} = -0.2700$$

$\implies$

$$\widehat{BOM} = 105.7^\circ = 1.844 \text{ rad}$$

and the arc is $r\theta = 6 \times 1.844 = \boxed{11.1}$ thousand km.

10 (f) $\mathbf{b} \times \mathbf{m} = (-15.15, -26.23, 16.84)$ and $\mathbf{b} \times \mathbf{p} = (-18, -31.18, 0)$, so

$$\cos \beta = \frac{1090.6}{34.66 \times 36} = 0.8740$$

$$\implies$$

$$\boxed{\beta = 029.1^\circ}$$

11 (a) $\overrightarrow{AB} = (5, -2, -1)$ and $\overrightarrow{ON} = (2 + 5\mu, -4 - 2\mu, 2 - \mu)$, so $\overrightarrow{PN} = (3 + 5\mu, -5 - 2\mu, 15 - \mu)$ and

$$\begin{aligned} \overrightarrow{PN} \cdot \overrightarrow{AB} &= 5(3 + 5\mu) \\ &\quad - 2(-5 - 2\mu) \\ &\quad - (15 - \mu) \\ &= \boxed{10 + 30\mu} \end{aligned}$$

11 (b) $10 + 30\mu = 0$ gives $\mu = -\frac{1}{3}$ and $N =$

$$\boxed{\left(\frac{1}{3}, -\frac{10}{3}, \frac{7}{3}\right)}.$$

11 (c) Here the given point is the origin, so the condition is $\overrightarrow{OP} \cdot \mathbf{d} = 0$ with $\overrightarrow{OP} = (\lambda, 2 - 2\lambda, 4 - 2\lambda)$:

$$\lambda$$

$$\begin{aligned} & -2(2-2\lambda) \\ & -2(4-2\lambda) = 9\lambda \\ & -12 = 0 \end{aligned}$$

$$\implies$$

$$\lambda = \frac{4}{3}$$

and $P = \boxed{\left(\frac{4}{3}, -\frac{2}{3}, \frac{4}{3}\right)}$.

12 (a) $\overrightarrow{PA} = (6, 0, 6)$, so $|\overrightarrow{PA}| = 6\sqrt{2}$ and

$$d = |\overrightarrow{PA}| \sin \frac{\pi}{3} = 6\sqrt{2} \cdot \frac{\sqrt{3}}{2} = \boxed{3\sqrt{6}} = \sqrt{54}$$

The product route agrees: $(6, 0, 6) \times (1, 1, 0) = (-6, 6, 6)$, of length $\sqrt{108}$, divided by $|(1, 1, 0)| = \sqrt{2}$.

12 (b) $(2, -2, 2) = 2(1, -1, 1)$, so the lines are parallel. With $A(3, 2, -1)$ on l_1 and $B(2, 0, 4)$ on l_2 , $\overrightarrow{AB} = (-1, -2, 5)$ and $\overrightarrow{AB} \times (1, -1, 1) = (3, 6, 3)$:

$$d = \frac{|(3, 6, 3)|}{|(1, -1, 1)|} = \frac{\sqrt{54}}{\sqrt{3}} = \sqrt{18} = \boxed{3\sqrt{2}}$$

Timer

$\mathbf{u} = (1, 1, 1)$ and $\mathbf{v} = (3, 1, -1)$. A vector perpendicular to both is along $\mathbf{u} \times \mathbf{v} = (-2, 4, -2)$, that is along $(1, -2, 1)$, so $\mathbf{w} = k(1, -2, 1)$.

Since $\mathbf{w} \perp \mathbf{v}$, the area of ACD is $\frac{1}{2}|\mathbf{v}||\mathbf{w}| = \frac{\sqrt{11}}{2}|\mathbf{w}| = 5$, giving $|\mathbf{w}| = \frac{10}{\sqrt{11}}$. As $|(1, -2, 1)| = \sqrt{6}$,

$$|k| = \frac{10}{\sqrt{66}} = \frac{5\sqrt{66}}{33} \approx 1.231$$

$$\mathbf{w} = \boxed{\pm \frac{5\sqrt{66}}{33} \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}} \approx \pm \begin{pmatrix} 1.23 \\ -2.46 \\ 1.23 \end{pmatrix}$$